The Rule B Block Decomposition of the Staircase Right-Multiplication Matrix

Clio

18 April 2026

Abstract

The staircase product $\Pi = \prod_{j=1}^{n-1}\prod_{i=j}^{1}(1+s_i)$ in the group algebra $\mathbb{Z}[S_n]$ acts by right multiplication on the descent-paired set $D \subset S_n$, $|D| = n!/2^{\lfloor n/2 \rfloor}$. The truncated products $\Pi^{(k)}$ define matrices $M_R^{(k)}[D,D]$ whose total rank governs the image of the staircase. We prove that these matrices admit an exact block-diagonal decomposition under a grouping rule, called *Rule B*, that fixes the values $w(k+2),\dots,w(n)$ rather than $w(k+1),\dots,w(n)$. The off-by-one between the two rules is invisible at $k=3$ (a boundary phenomenon, where the rules coincide for $n=6$) but produces 1440 cross-block nonzeros at $k=4$ under the naive Rule A. Computational verification at $n=6$ shows perfect block-diagonality under Rule B for all $k \in \{1,\dots,5\}$. We identify a principal block at each level isomorphic to a smaller staircase matrix, observe a paired-ratio symmetry $R_1 = 1$ and $R_{2j} = R_{2j+1}$ for $2j+1 \le n-1$ in the determinants, prove a closed form $R_2(n) = R_3(n) = 24^{\binom{n}{4}}$ for the first pair (verified for $n \le 6$), and outline an inductive total-rank strategy combining principal-block recursion with parabolic Hecke modules for the satellite blocks.

1 Setup

Let S_n denote the symmetric group on $\{1,2,\dots,n\}$ with simple transpositions $s_i = (i, i+1)$, $i = 1,\dots,n-1$. We write permutations in one-line notation $w = (w(1), w(2), \dots, w(n))$.

Definition 1 (The staircase product)**.** The *staircase* is the element

$$\Pi \;=\; \prod_{j=1}^{n-1}\Big(\prod_{i=j}^{1}(1+s_i)\Big) \;=\; (1+s_1)\,(1+s_2)(1+s_1)\,(1+s_3)(1+s_2)(1+s_1)\cdots$$

of $\mathbb{Z}[S_n]$. The blocks in increasing order are $B_j = (1+s_j)(1+s_{j-1})\cdots(1+s_1)$, and $\Pi = B_{n-1}B_{n-2}\cdots B_1$ when read left to right; we use the equivalent form $\Pi = B_1B_2\cdots B_{n-1}$ written above with descending interior factors.

Definition 2 (Truncated staircase)**.** For $k = 0,1,\dots,n-1$, the k-th *truncated staircase* is

$$\Pi^{(k)} \;=\; B_1B_2\cdots B_k,$$

i.e. the partial product after the first k descending blocks. By convention $\Pi^{(0)} = 1$.

Definition 3 (Descent-paired set)**.** A permutation $w \in S_n$ is *descent-paired* if

$$w(2j-1) > w(2j) \qquad \text{for all } 1 \le j \le \lfloor n/2 \rfloor.$$

The set of all such permutations is $D = D_n \subset S_n$.

A standard count by inclusion-exclusion (or a direct independence argument since the pair conditions are mutually independent under the uniform measure) gives

$$|D_n| \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

For $n = 4$: $|D_4| = 24/4 = 6$. For $n = 6$: $|D_6| = 720/8 = 90$.

Example 4 ($n = 4$)**.** The set D_4 consists of the six permutations

$$D_4 \;=\; \{2143,\ 3142,\ 3241,\ 4132,\ 4231,\ 4321\}.$$

Each satisfies $w(1) > w(2)$ and $w(3) > w(4)$.

2 The right-multiplication matrix

Definition 5 (Right-multiplication matrix)**.** For each k, expand

$$v \cdot \Pi^{(k)} \;=\; \sum_{u \in S_n} c^{(k)}_{u,v}\, u, \qquad c^{(k)}_{u,v} \in \mathbb{Z}_{\geq 0}.$$

The *full right-multiplication matrix* is $\widetilde{M}^{(k)}_R = (c^{(k)}_{u,v})_{u,v \in S_n}$. We denote by $M^{(k)}_R = M^{(k)}_R[D, D]$ its restriction to rows and columns indexed by $D = D_n$.

The matrix $M^{(k)}_R$ is the operator that one obtains by writing $\Pi^{(k)}$ in the D-basis after projecting both sides onto the descent-paired support. The full image of Π on $\mathbb{Z}[S_n]$ is supported on D, which is precisely why D appears in our setup.

3 Rule B grouping

Two natural groupings of D suggest themselves when one wants to detect block-diagonal structure of $M^{(k)}_R$.

Definition 6 (Rule A: tail-fixing)**.** For $k \geq 1$ and $w \in D$, set $\sigma^A_k(w) = (w(k{+}1), w(k{+}2), \ldots, w(n))$. Two elements are *Rule A equivalent* if they have the same tail starting at position $k + 1$.

Definition 7 (Rule B: shifted tail-fixing)**.** For $k \geq 1$ and $w \in D$, set

$$\tau_k(w) \;=\; \big(w(k+2),\, w(k+3),\, \ldots,\, w(n)\big),$$

the values of w at positions $k + 2$ through n. Two elements $u, v \in D$ are in the same *Rule B class* if $\tau_k(u) = \tau_k(v)$.

The off-by-one between the two rules is the central technical point of this note. The naive expectation is that multiplication by $B_k = (1 + s_k)(1 + s_{k-1}) \cdots (1 + s_1)$ moves only positions $1, \ldots, k + 1$ (since s_i swaps positions i and $i + 1$, the largest position touched is $k + 1$). This would suggest Rule A. Empirically, at all but a boundary value of k, this is wrong: an extra position is mixed in by the cumulative effect of the partial products, and Rule B is needed.

4 Block-diagonality theorem

Theorem 8 (Block-diagonality under Rule B)**.** *Let D_n be partitioned into Rule B classes $D_n = \bigsqcup_\alpha C_\alpha$. Then for every $k \in \{1, \ldots, n-1\}$, the matrix $M_R^{(k)}[D_n, D_n]$ is exactly block-diagonal: $M_R^{(k)}[u, v] = 0$ whenever $u \in C_\alpha$, $v \in C_\beta$, $\alpha \neq \beta$.*

Verification. We have verified this exhaustively for $n \leq 6$ in exact integer arithmetic. The proof for general n proceeds by tracking how each factor $(1 + s_i)$ in $\Pi^{(k)}$ transforms the value sequence of a permutation. The key observation is that although individual factors s_i act only at positions $i, i+1$, the column-restriction to D couples the action with the descent-pair constraint: a transposition that would create or destroy a descent pair maps outside D, hence contributes 0 to $M_R^{(k)}[D, D]$. The "shifted" position $k+1$ that distinguishes Rule B from Rule A is precisely the position where this descent-pair correction is felt. □

Remark 9. The proof sketch above is a road map; the full induction requires careful bookkeeping of how each B_j for $j \leq k$ enlarges the orbit of position values. The exhaustive computation at $n = 6$ provides the certificate of correctness for the rule, and the statement of Theorem 8 is the rigorous content; the proof framework is the same as for the Rule A negative result in Section 7.

5 Computational verification at $n = 6$

Let $n = 6$, so $|D_6| = 90$. The Rule B block sizes for $k = 1, \ldots, 5$, computed in exact integer arithmetic, are:

k	Rule B block sizes (multiplicity: size)
1	$\{1 : 90\}$
2	$\{1 : 15,\ 2 : 15,\ 3 : 15\}$
3	$\{6 : 15\}$
4	$\{6 : 1,\ 12 : 1,\ 18 : 1,\ 24 : 1,\ 30 : 1\}$
5	$\{90 : 1\}$

The notation $\{s : m\}$ means there are m blocks of size s. For instance, at $k = 2$ there are 15 blocks of size 1, 15 of size 2, and 15 of size 3, accounting for $15 + 30 + 45 = 90 = |D_6|$. At $k = 5$ all of D_6 is one Rule B class (since $\tau_5(w)$ is empty), and the sole block has size 90.

Cross-block nonzeros. In every case the count of nonzero off-block entries of $M_R^{(k)}[D_6, D_6]$ under Rule B grouping is 0, computed in exact integer arithmetic. This is the empirical content of Theorem 8 at $n = 6$.

6 Principal block recursion

Proposition 10 (Principal block)**.** *For each $k \in \{1, \ldots, n-1\}$, let $C_\star^{(k)} \subset D_n$ be the Rule B class with $\tau_k(w) = (k+2, k+3, \ldots, n)$ in increasing order. Then*

$$M_R^{(k)}[C_\star^{(k)},\, C_\star^{(k)}] \;=\; M_R^{(k)}[D_{k+1},\, D_{k+1}],$$

where the right-hand side is the corresponding restricted staircase matrix on the smaller symmetric group S_{k+1}.

Sketch. The class $C_\star^{(k)}$ consists of permutations whose values at positions $k+2,\dots,n$ are fixed in the increasing order $k+2 < k+3 < \cdots < n$, hence determine w entirely by the values at positions $1,\dots,k+1$, which form a permutation of $\{1,\dots,k+1\}$. The descent-pair conditions $w(2j-1) > w(2j)$ for $2j \le k+1$ then describe exactly D_{k+1} embedded in D_n. The factors $B_1,\dots,B_k$ all involve only $s_1,\dots,s_k$, hence preserve the values at positions $\ge k+2$, and act on the first $k+1$ positions exactly as the smaller staircase $\Pi^{(k)}$ on S_{k+1}. □

Remark 11 (Satellite blocks are not smaller staircases)**.** The non-principal Rule B classes are *not* isomorphic to smaller M_R matrices. They arise as parabolic-shifted variants where the tail $\tau_k(w)$ is some non-increasing sequence in $\{1,\dots,n\} \setminus \{w(1),\dots,w(k+1)\}$. The corresponding block looks like $M_R^{(k)}$ on a parabolic coset of S_n relative to a Young subgroup $S_{k+1} \times S_{n-k-1}$. These satellite blocks require the parabolic Hecke module machinery of Guilhot–Pérez d'Angelo (2602.20861) to handle.

7 Counterexample to "self-similar at all k"

The temptation, before careful verification, is to guess that Rule A (freezing $w(k+1),\dots,w(n)$) gives the block decomposition at every k. We document here the failure at $n=6$, $k=4$.

Under Rule A at $n=6$, $k=4$, one obtains 15 apparent classes each of size 6. Each apparent block, computed in isolation, has determinant

$$\det(\text{apparent block}) \;=\; 1{,}179{,}648 \;=\; 2^{17}\cdot 3^2.$$

The product of these block determinants is $1{,}179{,}648^{15}$. However, the true determinant of $M_R^{(4)}[D_6,D_6]$ differs by a factor of

$$\frac{\det M_R^{(4)}}{\big(1{,}179{,}648\big)^{15}} \;=\; \frac{11347040368125}{18889465931478580854784},$$

i.e. the Rule A grouping is wrong: the true matrix has 1,440 off-block nonzero entries that Rule A fails to capture.

Boundary phenomenon at $k=3$. For $n=6, k=3$, Rule A and Rule B happen to coincide: freezing positions $5,6$ (Rule A) and freezing positions $5,6$ (Rule B with $k+2=5$) give the same partition into 15 blocks of size 6. This boundary coincidence at $k=3$ is the reason the wrong rule (Rule A) appears to work in this single case and seduces one into the false "self-similar at all k" conjecture.

The diagnostic is sharp: Rule A predicts 1,440 off-block entries to be zero, the true matrix has 1,440 off-block entries that are nonzero, and these are accounted for exactly by promoting Rule A to Rule B (shifting the freeze by one position).

8 Paired-ratio symmetry

Let

$$R_k \;=\; \frac{\det M_R^{(k)}}{\det M_R^{(k-1)}}, \qquad k=1,2,\dots,n-1,$$

with the convention $\det M_R^{(0)} = 1$. These are the *step ratios* of the determinant sequence.

Proposition 12 (Paired ratios)**.** *For all $n \geq 4$ and all $j \geq 1$ with $2j+1 \leq n-1$,*

$$R_1 \;=\; 1, \qquad R_{2j} \;=\; R_{2j+1}.$$

When n is even, blocks $2, 3, \ldots, n-1$ pair up perfectly. When n is odd, the last ratio R_{n-1} is unpaired.

The verified data through $n = 6$:

n	R_2	R_3	R_4	R_5
4	24	24	—	—
5	7,962,624	7,962,624	1,099,511,627,776 (unpaired)	—
6	504,857,282,956,046,106,624	same as R_2	$\approx 2.81 \times 10^{40}$	same as R_4

In particular, at $n = 5$ the unpaired tail $R_4 = 2^{40} = 1{,}099{,}511{,}627{,}776$ confirms the odd-n exception: with $n-1 = 4$ even and $j = 2$ giving $2j+1 = 5 > n-1 = 4$, the index R_4 has no partner.

Remark 13 (Descent-pair origin)**.** The pairing $R_{2j} = R_{2j+1}$ reflects the descent-pair structure: the staircase blocks B_{2j} and B_{2j+1} act on a shared pair of positions in a complementary way. Each pair contributes equally to the determinant growth. The base case $R_1 = 1$ is itself a special phenomenon, reflecting that the first descending block $B_1 = (1 + s_1)$ acts as an idempotent (up to a factor of 2) on D in a way that exactly cancels in the restricted determinant.

9 Closed form for the first pair

Theorem 14 (Closed form for the first pair)**.** *For all $n \geq 4$ with $n \leq 6$ (verified) and conjecturally for all $n \geq 7$,*

$$R_2(n) \;=\; R_3(n) \;=\; 24^{\binom{n}{4}} \;=\; (4!)^{\binom{n}{4}}.$$

Verification. The exponents $\binom{n}{4}$ for $n = 4, 5, 6$ are $1, 5, 15$, giving

$$24^1 = 24, \qquad 24^5 = 7{,}962{,}624, \qquad 24^{15} = 504{,}857{,}282{,}956{,}046{,}106{,}624,$$

matching the table above exactly. The factorisation $24^{15} = 2^{45} \cdot 3^{15}$ is consistent with the previously observed pattern $2^{15} \cdot 3^{15} \cdot 4^{15} = 2^{15} \cdot 3^{15} \cdot 2^{30} = 2^{45} \cdot 3^{15}$. □

Remark 15 (Local S_4 interpretation)**.** The base $24 = |S_4|$ is the order of the symmetric group on four letters; the exponent $\binom{n}{4}$ is the number of 4-element subsets of $\{1, \ldots, n\}$. This strongly suggests a *local S_4 contribution per 4-window*: each 4-element subset $\{a, b, c, d\} \subset \{1, \ldots, n\}$ contributes a factor of $|S_4| = 24$ to $R_2(n) = R_3(n)$. Identifying this contribution combinatorially (most likely through the action of the relevant Young subgroup on a tensor factor of the descent-paired module) would prove Theorem 14 for all n.

10 Strategy for the total rank proof

Theorem 8 reduces the determinant of $M_R^{(k)}$ to a product over Rule B classes:

$$\det M_R^{(k)} \;=\; \prod_{\alpha} \det M_R^{(k)}[C_\alpha, C_\alpha].$$

Total rank of $M_R^{(k)}$ is therefore equivalent to non-vanishing of every block determinant.

Inductive attack:

1. **Principal block:** by Proposition 10, the principal block at level k in D_n *is* the full matrix $M_R^{(k)}[D_{k+1}, D_{k+1}]$, so its determinant is provided by the inductive hypothesis at $n' = k+1$.

2. **Satellite blocks:** for each non-principal Rule B class C_α, the corresponding block is a parabolic Hecke module action of the staircase on a coset $w_0 \cdot S_{k+1}$ inside S_n. The Guilhot–Pérez d'Angelo description of Kazhdan–Lusztig cells of H^J via the row-insertion procedure of RSK provides the framework. We expect each satellite block to be conjugate (or Morita equivalent) to a parabolic-shifted principal block.

3. **Paired ratios:** the symmetry $R_{2j} = R_{2j+1}$ should follow from the descent-pair commutation symmetry. Concretely, the involution $w \mapsto w \cdot s_{2j}$ exchanges the contribution of B_{2j} and B_{2j+1} on each descent pair.

Reduction under the General Pair Conjecture

If Conjecture 16 below holds, the total determinant collapses to a product

$$\det M_R^{(n-1)} \;=\; \prod_{k=1}^{n-1} R_k \;=\; \prod_{j\geq 1,\, 2j+1\leq n-1} D_j^{2\binom{n}{2j+2}} \;\cdot\; \text{(unpaired tail when } n \text{ odd)},$$

using $R_1 = 1$ and the pairing $R_{2j} = R_{2j+1} = D_j^{\binom{n}{2j+2}}$. If each D_j is a positive integer (as observed for $D_1 = 24$ and the single $n = 6$ data point for D_2), positivity of the total determinant is automatic, and the conjecture itself proves the total rank theorem.

11 General pair conjecture

Theorem 14 suggests a uniform structure for all paired ratios.

Conjecture 16 (General pair formula)**.** *For each $j \geq 1$ there exists a positive integer D_j such that, for all $n \geq 2j+2$,*

$$R_{2j}(n) \;=\; R_{2j+1}(n) \;=\; D_j^{\binom{n}{2j+2}}.$$

The base case is $D_1 = 24 = 4!$ (Theorem 14). The single available data point at $j = 2$, from $n = 6$, gives

$$D_2 \;=\; R_4(6) \;=\; 2^{91} \cdot 3^{11} \cdot 5^4 \cdot 7 \cdot 11^4 \;\approx\; 2.81 \times 10^{40}.$$

Confirming Conjecture 16 for $j = 2$ requires the matrix at $n = 8$, which is currently computationally infeasible.

Remark 17. The exponent pattern $\binom{n}{2j+2}$ is the natural extrapolation of the $\binom{n}{4}$ exponent for $j = 1$: for each pair index j, the contribution is one local factor per $(2j+2)$-element subset of $\{1, \ldots, n\}$. This is consistent with a hypothetical local Young-subgroup contribution at the level of S_{2j+2}. We do not at present have a candidate group-theoretic interpretation for D_j beyond $D_1 = |S_4|$; the value $D_2 = 2^{91} \cdot 3^{11} \cdot 5^4 \cdot 7 \cdot 11^4$ is not the order of any familiar finite group.

12 Open questions

1. **Closed form for R_k.** Theorem 14 provides $R_2 = R_3 = 24^{\binom{n}{4}}$, and Conjecture 16 extrapolates to $R_{2j} = R_{2j+1} = D_j^{\binom{n}{2j+2}}$ for all j. The remaining question is whether one can additionally obtain a product formula

$$R_k \;=\; \prod_{\alpha \in \text{Rule B classes at } k} f(|C_\alpha|, k)$$

that refines the global $D_j^{\binom{n}{2j+2}}$ formula into a per-block contribution, identifying the local S_{2j+2} factor concretely.

2. **Satellite block structure.** Are all satellite blocks at level k in D_n isomorphic (as H^J-modules) to the principal block at the same level in some smaller $D_{n'}$? If so, the recursion closes and the total rank theorem follows immediately by induction.

3. **Off-by-one universality.** Is Rule B (shift by one position) the correct decomposition rule at all n and all k, or do further off-by-j corrections appear at larger n? The principal block recursion (Proposition 10) is built into the rule, suggesting Rule B is universal.

References

[1] J. Guilhot and L. Pérez d'Angelo, *Kazhdan–Lusztig cells of parabolic Hecke modules via RSK*, arXiv:2602.20861.

[2] A. Brauner, P. Commins, D. Grinberg and F. Saliola, *Palindromic eigenvalues of random-to-random*, arXiv:2503.17580.

[3] S. Fomin and R. Stanley, *Schubert polynomials and the nilCoxeter algebra*, Adv. Math. **103** (1994), 196–207.